Proposal Outline: DUE March 27th by Noon

(LastName_Assignment6.doc on D2L)

1) Revise Title

2) Revise abstract.

Adhere to word limit: The abstract must be written in standard ASCII and should be no longer than 20 lines of 85 characters of text. Use the HST/JWST Templates.

Science Justification will need to be 3 pages in total. This includes 1 figure, but not the bibliography.

<u>**Science Justification Outline**</u>

3) Provide an outline for your proposal (BULLETED POINTS that answer each of the below)

<u>**Section 1:** Facts, Problem & Summary paragraph (3/4 of a page)</u>

a) What is the descriptive heading? It must be a *statement*

b) 1 fact-based paragraph:
   i) State the fact leading the paragraph (the conclusion) – this should follow from the abstract.
   ii) What would be supporting facts (1-3)? Provide at least one paper citation per supporting fact.
   iii) What is the nuanced conclusion for the paragraph? This is the importance of the fact that should set the stage for the problem (should follow from abstract)

c) Problem paragraph:
   iv) What is "the problem" statement?
   v) What references are needed to support the problem?
   vi) Why is the problem critical to answer now? (nuanced conclusion)

d) Synopsis paragraph:
   vii) What is the proposal? (identify key component) "We propose"
   viii) Why the facility? This can be related to the funding call.
   ix) What is the broader impact ? Community deliverables?

<u>**Section 2:** The Problem & Motivation/Narrative of the proposal</u>

a) Identify a descriptive heading that <u>states</u> the problem
b) What fact do you need to start with to set up the problem?
c) What references support the problem?
d) State why solving this problem advances the subfield.

<u>**Section 3:** The Strategy: HST/JWST + Target Solves the Problem (Generate the "Key Component")</u>

a) Create a descriptive heading that <u>states</u> the strategy

b) What are you targeting ? (imaging/spectra/archival data of **what astronomical object**, or **what physical mechanism/ simulations/data set**)
c) Why this facility? (Suitability in more detail)
d) Outline the strategy to generate the key component
For GO: describe the proposed observations *briefly/succinctly*
For AR/Theory: describe *briefly* the analysis method/proposed usage of simulations/codes and relation to existing/future HST/JWST data sets.
e) Outline the products of the program (tangible outcomes) and state that these products solve the problem.

Include a Figure that explains the strategy/target/solution
Add here your (revised) Figure from Assignment 4 (or you must create a new one if your proposal is different)
Figure and caption instructions are in Assignment 4.

Section 4: Importance and Broader Implications
a) Create a heading that states the broader impact/importance of the solution.
b) What is the ***high level*** importance of the proposed products of your study to your **subfield** ? How will your study advance the subfield? (this is a like a summary statement)
c) Repeat Suitability:
   i) Why now?
   ii) How does this program relate to the funding call?
      Which **key science goals** of HST/JWST does your proposal connect to ?
      - JWST: https://webb.nasa.gov/content/science/index.html
      - HST: https://www.stsci.edu/hst/about/key-science-themes
      [if applicable -> Observational] Repeat Why this facility?

d) How does this study increase the **legacy value** of the facility/agency?
   - What **other subfields** are advanced by these observations (GO) or data analysis/code/simulations (AR) ? [Broader Impact]

   And/OR
   - Do the products of your proposal **serve the community** at large?